

Module 3: Data Link Layer and Wireless Networks



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Data-Link Layer

- Data Link Control (DLC)
- Multiple Access Protocols (MAC)
- Link-Layer Addressing
- Ethernet Protocol
- Connecting Devices

Wireless Networks

- Wireless LANs
- Mobile IP

Hands-On

- Datalink Provider Interface
- SOCK_PACKET
- PF_PACKET

Multiple Access Protocols (MAC)

When multiple stations share the same communication medium, a protocol is required to coordinate access.

Examples:

- Wireless LANs
- Cellular Networks
- Shared Ethernet Networks



Goal

Prevent collisions and ensure fair access to the shared medium.

Need for Multiple Access Protocols

In a shared medium:

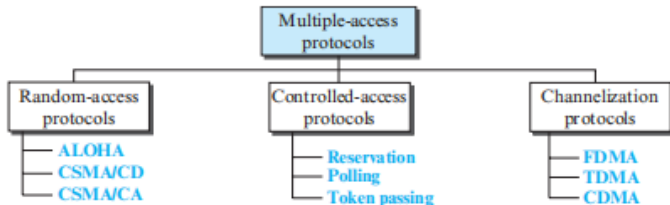
- Multiple stations may transmit simultaneously.
- Simultaneous transmissions cause collisions.
- Collisions destroy transmitted frames.

MAC protocols help:

- Avoid collisions
- Detect collisions
- Recover from collisions

Categories of Multiple Access Protocols

Figure 5.28 *Taxonomy of multiple-access protocols discussed in this chapter*



Multiple access protocols are classified into:

- ① Random Access
- ② Controlled Access
- ③ Channelization

This section focuses on Random Access protocols.

Random Access Protocols

Random Access is also called:

- Contention Method

Characteristics:

- No central controller.
- All stations have equal priority.
- Stations compete for medium access.
- Transmission occurs whenever a station has data.

Questions Addressed by Random Access

Every station must answer:

- 1 When can the medium be accessed?
- 2 What if the medium is busy?
- 3 How is transmission success detected?
- 4 What happens after a collision?

Protocols define rules to answer these questions.

Evolution of Random Access Methods

ALOHA



CSMA



CSMA/CA

Each new protocol improves efficiency and reduces collisions.

ALOHA Protocol

ALOHA was developed at the University of Hawaii in the early 1970s.

Characteristics:

- Earliest random access protocol.
- Designed for wireless communication.
- Can be used on any shared medium.

Basic Idea:

Transmit whenever a station has data.

Pure ALOHA

Pure ALOHA follows a very simple rule:

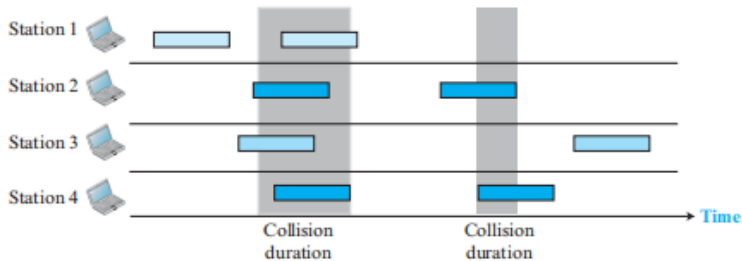
A station transmits immediately whenever it has a frame.

Problem:

- Multiple stations may transmit simultaneously.
- Frame collisions can occur.

Frame Collisions in Pure ALOHA

Figure 5.29 Frames in a pure ALOHA network



Observations:

- Several frames collide.
- Collided frames are destroyed.
- Only non-overlapping frames survive.

Operation of Pure ALOHA

Steps:

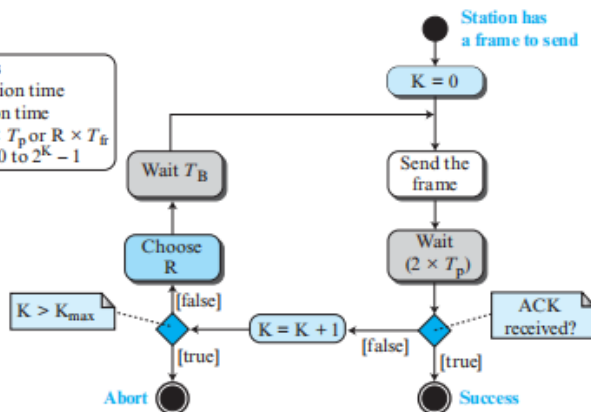
- 1 Send frame immediately.
- 2 Wait for acknowledgment.
- 3 If ACK arrives, transmission succeeds.
- 4 If timeout occurs, assume collision.
- 5 Retransmit after random backoff time.

Pure ALOHA Algorithm

Figure 5.30 Procedure for pure ALOHA protocol

Legend

K : Number of attempts
 T_p : Maximum propagation time
 T_{fr} : Average transmission time
 T_B : (Back-off time): $R \times T_p$ or $R \times T_{fr}$
 R : (Random number): 0 to $2^K - 1$



Pure ALOHA Algorithm

Key Concepts:

- Timeout period
- Retransmission
- Random Backoff
- Maximum Retry Limit

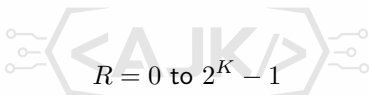


Binary Exponential Backoff

Backoff Time:

$$T_B = R \times T_p$$

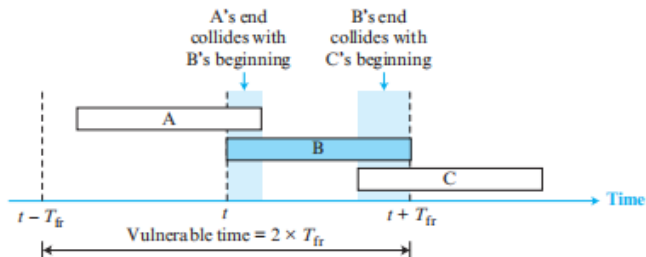
where


$$R = 0 \text{ to } 2^K - 1$$

- K = Number of failed transmissions
- Backoff range increases after every collision

Vulnerable Time in Pure ALOHA

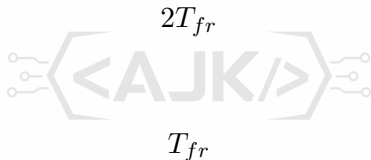
Figure 5.31 *Vulnerable time for pure ALOHA protocol*



Vulnerable Time in Pure ALOHA

Vulnerable Time:

where


$$2T_{fr}$$
$$T_{fr}$$

is frame transmission time.

Any frame arriving during this interval causes collision.

Throughput of Pure ALOHA

Throughput Equation:

$$S = Ge^{-2G}$$

where

- S = Throughput
- G = Offered Load

Maximum Throughput:

$$S_{max} = 0.184$$

at

$$G = \frac{1}{2}$$

Only 18.4% efficiency.

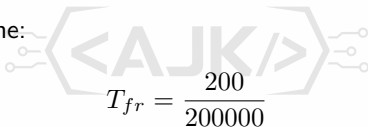
Pure ALOHA Example

Given:

$$\text{Frame} = 200 \text{ bits}$$

$$\text{Bandwidth} = 200 \text{ kbps}$$

Frame Transmission Time:


$$T_{fr} = \frac{200}{200000}$$

$$T_{fr} = 1 \text{ ms}$$

Vulnerable Time:

$$2T_{fr} = 2 \text{ ms}$$

Slotted ALOHA

Improvement over Pure ALOHA.

Idea:

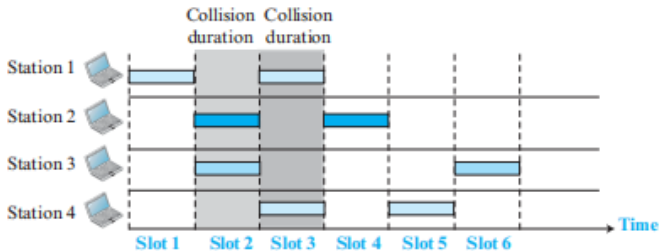
- Divide time into slots.
- Station can transmit only at slot boundaries.

Result:

- Fewer collisions.
- Better throughput.

Frame Collisions in Slotted ALOHA

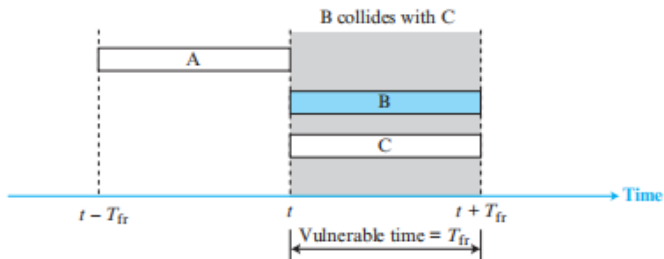
Figure 5.32 *Frames in a slotted ALOHA network*



Collisions occur only when two stations choose the same slot.

Vulnerable Time in Slotted ALOHA

Figure 5.33 Vulnerable time for slotted ALOHA protocol



Vulnerable Time:

$$T_{fr}$$

Only one frame transmission time.

This is half the vulnerable period of Pure ALOHA.

Pure ALOHA vs Slotted ALOHA

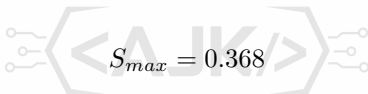
Feature	Pure	Slotted
Transmission Time	Anytime	Slot Boundaries
Synchronization	Not Required	Required
Vulnerable Time	$2T_{fr}$	T_{fr}
Maximum Throughput	18.4%	36.8%
Efficiency	Lower	Higher

Throughput of Slotted ALOHA

Throughput Equation:

$$S = Ge^{-G}$$

Maximum Throughput:


$$S_{max} = 0.368$$

at

$$G = 1$$

Efficiency doubles compared to Pure ALOHA.

Key Takeaways

- Multiple Access Protocols manage shared media.
- ALOHA is the earliest random-access protocol.
- Pure ALOHA allows transmission anytime.
- Slotted ALOHA uses synchronized slots.
- Slotted ALOHA doubles throughput.
- Vulnerable time decreases from $2T_{fr}$ to T_{fr} .

Example 5.14: Slotted ALOHA Throughput

$$T_{fr} = \frac{200}{200000} = 1ms$$

$$S = Ge^{-G}$$

Frames/s	G	$S=Ge^{-G}$	Throughput
1000	1	0.368	368
500	0.5	0.303	151
250	0.25	0.195	49

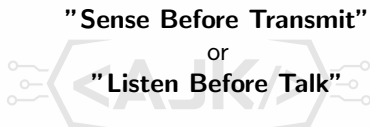
Observation

Maximum throughput occurs when $G = 1$, giving $S_{max} = 0.368$ (36.8%).

Carrier Sense Multiple Access (CSMA)

- Developed to reduce collisions and improve channel utilization.
- Before transmitting, a station listens to the medium.
- Follows the principle:

"Sense Before Transmit"
or
"Listen Before Talk"



- If the medium is idle, the station may transmit.
- If the medium is busy, the station follows a persistence rule.
- CSMA reduces collisions but cannot eliminate them completely.

Key Idea

A station checks the channel status before sending data.

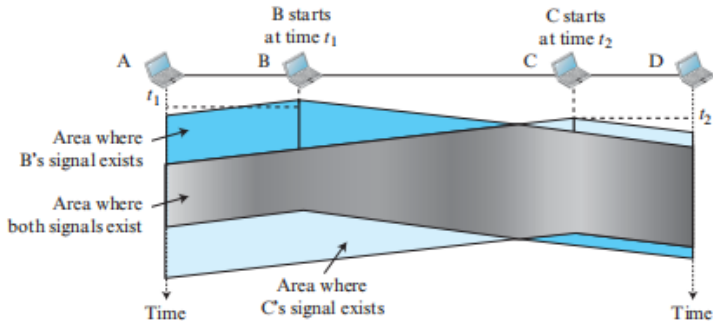
Why Collisions Still Occur in CSMA

- Collision is still possible because of **propagation delay**.
- A station may sense the channel as idle before another station's signal reaches it.
- Example:
 - Station B starts transmitting at time t_1 .
 - Station C senses the channel at time t_2 .
 - Signal from B has not yet reached C.
 - C also starts transmitting.
 - Frames collide and are destroyed.

Important

Carrier sensing reduces collisions but cannot prevent collisions caused by propagation delay.

Figure 5.34 *Spacetime model of a collision in CSMA*



Vulnerable Time in CSMA

- Vulnerable Time = Propagation Time (T_p)
- If another station transmits during T_p , a collision occurs.
- Once the first bit reaches all stations, every station becomes aware of the transmission.
- No new transmission will begin until the channel becomes idle again.

Formula

$$\text{Vulnerable Time} = T_p$$

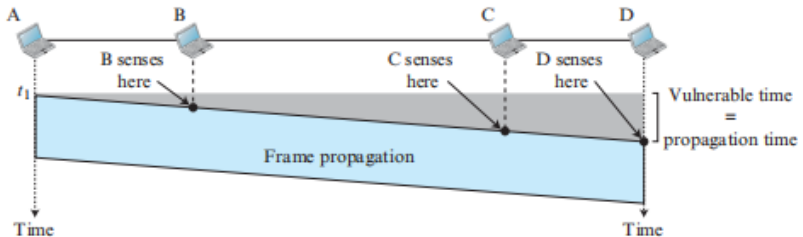
Comparison

Pure ALOHA : $2T_{fr}$

Slotted ALOHA : T_{fr}

CSMA : T_p

Figure 5.35 Vulnerable time in CSMA



Persistence Methods in CSMA

Method	Behavior
1-Persistent	Transmit immediately when channel becomes idle. High collision probability.
Nonpersistent	Wait a random time if channel is busy, then sense again. Fewer collisions but lower efficiency.
p-Persistent	When channel is idle: • Send with probability p • Wait one slot with probability $(1 - p)$ Provides balance between efficiency and collisions.

Figure 5.36 Behavior of three persistence methods

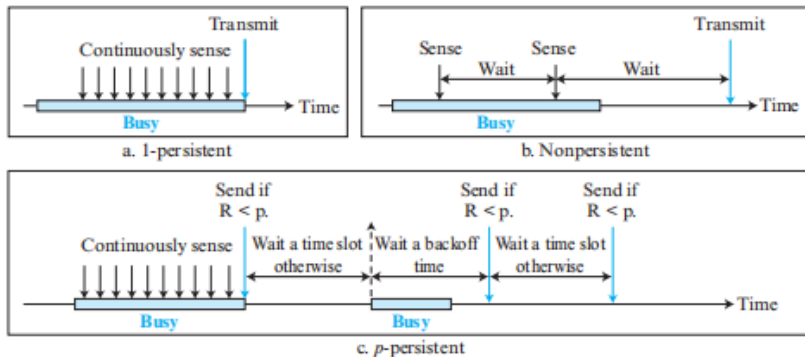
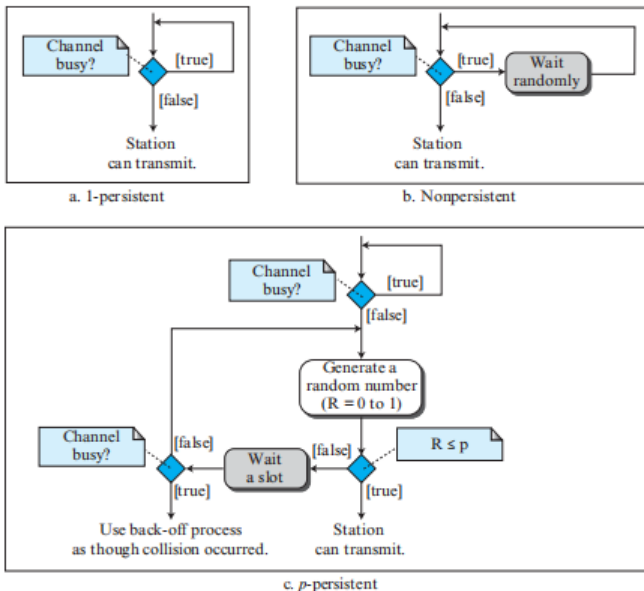


Figure 5.37 Flow diagram for three persistence methods



CSMA/CD (Carrier Sense Multiple Access with Collision Detection)

- CSMA reduces collisions but cannot eliminate them.
- CSMA/CD extends CSMA by detecting collisions after transmission begins.
- A station:
 - 1 Senses the channel.
 - 2 Transmits if the channel is idle.
 - 3 Continues monitoring the channel while transmitting.
 - 4 If no collision occurs, transmission succeeds.
 - 5 If a collision occurs, transmission is aborted and retransmitted later.

Key Idea

Listen before transmission and monitor during transmission.

Collision Detection in CSMA/CD

- Station A begins transmission at time t_1 .
- Before A's signal reaches C, station C senses the channel and starts transmitting at time t_2 .
- Signals collide in the medium.
- Station C detects collision at t_3 and stops transmission.
- Station A detects collision at t_4 and stops transmission.

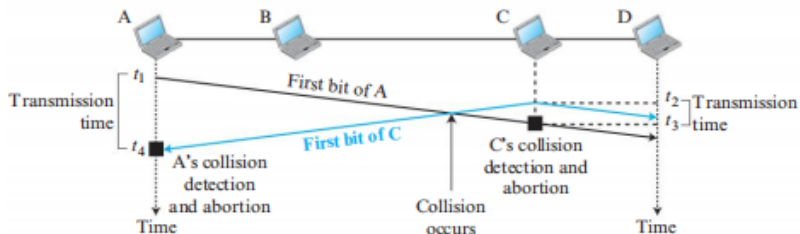
$$\text{Transmission Time of A} = t_4 - t_1$$

$$\text{Transmission Time of C} = t_3 - t_2$$

Collision Handling

Both stations abort transmission and retry after a backoff period.

Figure 5.38 Collision of the first bits in CSMA/CD



Minimum Frame Size in CSMA/CD

For collision detection to work:

$$T_{fr} \geq 2T_p$$

where,

- T_{fr} = Frame Transmission Time
- T_p = Maximum Propagation Time

Reason:

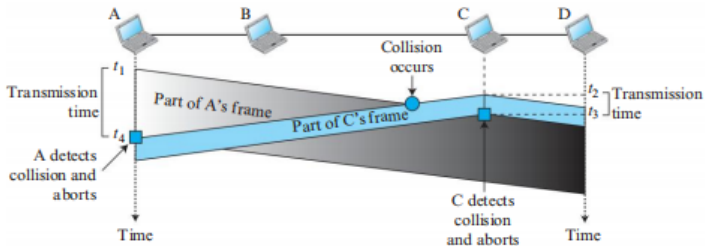
- A sender must still be transmitting when a collision signal returns.
- If transmission finishes before collision detection, the collision will go unnoticed.
- Therefore, frame size must be large enough to ensure collision detection.

Key Point

Minimum frame transmission time must be at least twice the maximum propagation delay.

Insert Figure 5.39 here

Figure 5.39 Collision and abortion in CSMA/CD

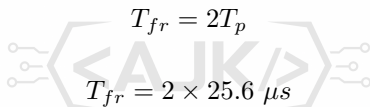


Example 5.15: Minimum Frame Size in CSMA/CD

Given:

- Bandwidth = 10 Mbps
- Maximum Propagation Time (T_p) = $25.6 \mu s$

Step 1: Find Minimum Frame Transmission Time

$$T_{fr} = 2T_p$$

$$T_{fr} = 2 \times 25.6 \mu s$$

$$T_{fr} = 51.2 \mu s$$

Step 2: Calculate Minimum Frame Size

$$\text{Frame Size} = \text{Bandwidth} \times T_{fr}$$

$$= 10 \times 10^6 \times 51.2 \times 10^{-6}$$

CSMA/CD Procedure

CSMA/CD follows these major steps:

- ① Sense the channel.
- ② Apply a persistence method.
- ③ If channel is idle, begin transmission.
- ④ Continuously monitor the medium.
- ⑤ If transmission completes successfully, stop.
- ⑥ If collision occurs:
 - Abort transmission.
 - Send a jamming signal.
 - Wait for a backoff time.
 - Retransmit later.

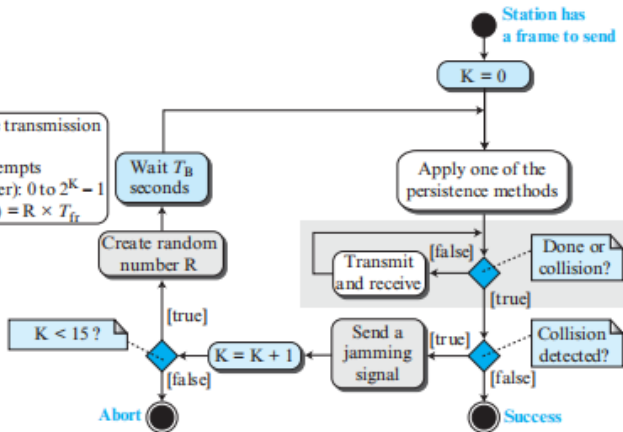
Key Difference from ALOHA

Collision detection occurs during transmission, not after the entire frame is sent.

Figure 5.40 Flow diagram for the CSMA/CD

Legend

T_{fr} : Frame average transmission time
 K : Number of attempts
 R : (random number): 0 to $2^K - 1$
 T_B : (Back-off time) = $R \times T_{fr}$



CSMA/CD Transmission Process

- A station continuously transmits and listens at the same time.
- Two possible events can occur:
 - 1 Transmission completed successfully.
 - 2 Collision detected.
- If transmission finishes:
 - Frame is successfully delivered.
- If collision is detected:
 - Transmission is immediately aborted.
 - A jamming signal is sent.
 - Station enters backoff state.

Important

The station does not wait until the end of the frame to detect collisions.

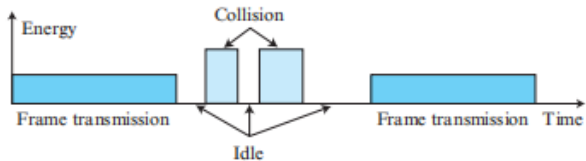
Energy Levels in CSMA/CD

A station monitors the channel energy level.

Energy Level	Meaning
Zero	Channel is idle
Normal	Successful transmission in progress
Abnormal	Collision detected

- Collision causes energy level to become approximately twice the normal level.
- Stations use energy monitoring to detect collisions.

Figure 5.41 *Energy level during transmission, idleness, or collision*



Throughput of CSMA/CD

- CSMA/CD provides much higher throughput than ALOHA.
- Throughput depends on:
 - Persistence method
 - Traffic load (G)
 - Network conditions

Method	Maximum Throughput
Pure ALOHA	18.4%
Slotted ALOHA	36.8%
1-Persistent CSMA/CD	Around 50%
Nonpersistent CSMA/CD	Up to 90%

Traditional Ethernet

Traditional Ethernet (10 Mbps) used **1-persistent CSMA/CD**.

CSMA/CA (Collision Avoidance)

- CSMA/CA stands for **Carrier Sense Multiple Access with Collision Avoidance**.
- It is primarily used in **Wireless LANs (IEEE 802.11 Wi-Fi)**.
- Unlike CSMA/CD, collision detection is difficult in wireless networks.
- Therefore, wireless stations try to avoid collisions before transmission.
- A station senses the channel before transmitting.
- If the channel is busy, the station waits for a random backoff period.
- If the channel is idle, the station transmits.

Key Idea

CSMA/CD detects collisions after they occur, while CSMA/CA attempts to avoid collisions before transmission.

CSMA/CD vs CSMA/CA

Feature	CSMA/CD	CSMA/CA
Full Form	Collision Detection	Collision Avoidance
Network Type	Wired LAN	Wireless LAN
Collision Handling	Detect and Abort	Avoid Before Sending
Medium Monitoring	During Transmission	Before Transmission
Used In	Traditional Ethernet	Wi-Fi (802.11)

Remember

Ethernet uses CSMA/CD, while Wireless LANs use CSMA/CA.

Controlled Access

- In controlled access, stations coordinate with each other before transmission.
- A station can transmit only after obtaining permission.
- Goal: Eliminate collisions and ensure fair channel access.
- Common Controlled Access Methods:
 - Reservation
 - Polling
 - Token Passing

Key Idea

Unlike random access methods, controlled access grants transmission rights explicitly.

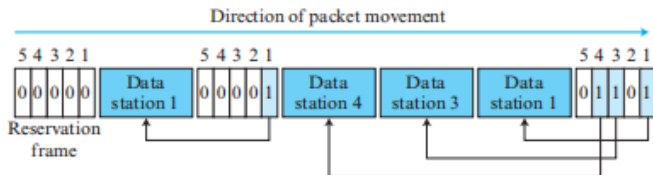
Reservation Method

- A station reserves the channel before transmitting data.
- Time is divided into intervals.
- Each interval begins with a Reservation Frame.
- Reservation frame contains **N minislots** for N stations.
- Each station owns one minislot.
- A station marks its minislot when it has data to send.
- Only stations with reservations transmit in the data phase.

Advantage

Collisions are avoided because transmission rights are reserved beforehand.

Figure 5.42 *Reservation access method*



Polling Method

- Uses a **Primary Station** and multiple **Secondary Stations**.
- Primary station controls the communication channel.
- Secondary stations transmit only when permitted.
- Prevents collisions through centralized control.

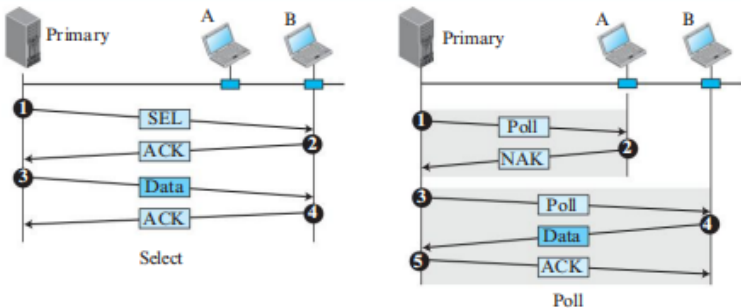
Polling Operations

- **Select:** Primary sends data to a secondary station.
- **Poll:** Primary asks secondary stations whether they have data to send.

Limitation

Failure of the primary station causes the entire system to stop.

Figure 5.43 *Select and poll functions in polling-access method*



Select Operation

- Used when the primary station wants to send data.
- Primary checks whether the target secondary is ready.
- Primary sends a **Select (SEL) Frame**.
- SEL frame contains the address of the intended secondary station.
- Secondary replies indicating readiness.
- After acknowledgment, data transmission begins.

Purpose

Ensures the receiver is ready before transmission starts.

Poll Operation

- Used when the primary station is ready to receive data.
- Primary polls secondary stations one by one.
- Secondary responds with:
 - NAK → No data available
 - Data Frame → Data ready for transmission
- Primary acknowledges received data using ACK.

Purpose

Allows secondary stations to transmit without causing collisions.

Token Passing

- Stations are organized into a logical ring.
- A special packet called a **Token** circulates continuously.
- Possession of the token grants transmission rights.
- Station transmits only while holding the token.
- After transmission, token is passed to the successor station.
- If no data is available, token is immediately forwarded.

Advantage

Collision-free channel access with fair sharing among stations.

Token Management

- Controls circulation of the token.
- Prevents a station from holding the token indefinitely.
- Detects lost or damaged tokens.
- Handles station failures while holding the token.
- Supports priority-based access.
- Allows high-priority stations to obtain the token sooner.

Need for Token Management

Ensures fairness, reliability, and efficient utilization of the network.

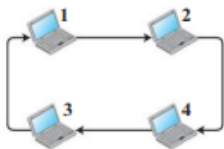
Logical Ring Topologies

Token-passing networks may use different physical topologies while maintaining a logical ring.

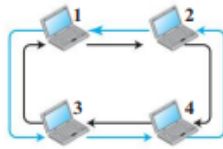
- Physical Ring
- Dual Ring
- Bus Ring (Token Bus)
- Star Ring (Token Ring)



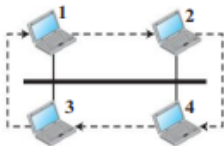
Figure 5.44 *Logical ring and physical topology in token-passing access method*



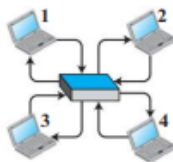
a. Physical ring



b. Dual ring



c. Bus ring



d. Star ring

Physical Ring and Dual Ring

Physical Ring

- Stations connected in a circular fashion.
- Token travels directly to successor station.
- Failure of one link may disrupt the entire network.

Dual Ring

- Uses a secondary backup ring.
- Backup ring operates in reverse direction.
- Provides fault tolerance and reliability.
- Used in FDDI and CDDI networks.

Bus Ring and Star Ring

Bus Ring (Token Bus)

- Physical topology is a bus.
- Stations logically form a ring.
- Token contains successor's address.
- Used in IEEE Token Bus LANs.

Star Ring (Token Ring)

- Physical topology is a star.
- Central hub internally creates the logical ring.
- Easier fault handling and station management.
- Used in IBM Token Ring networks.

Channelization

- Channelization divides the available bandwidth among users.
- Also known as **Channel Partitioning**.
- Multiple stations share the medium simultaneously.
- Resources are divided by:
 - Time
 - Frequency
 - Code
- Widely used in wireless communication systems.



Examples

FDMA, TDMA, and CDMA

Controlled Access vs Random Access

Feature	Random Access	Controlled Access
Transmission Permission	Not Required	Required
Collision Probability	High	Very Low
Coordination	Minimal	Explicit Coordination
Examples	ALOHA, CSMA	Reservation, Polling, Token Passing
Efficiency Under Load	Lower	Higher